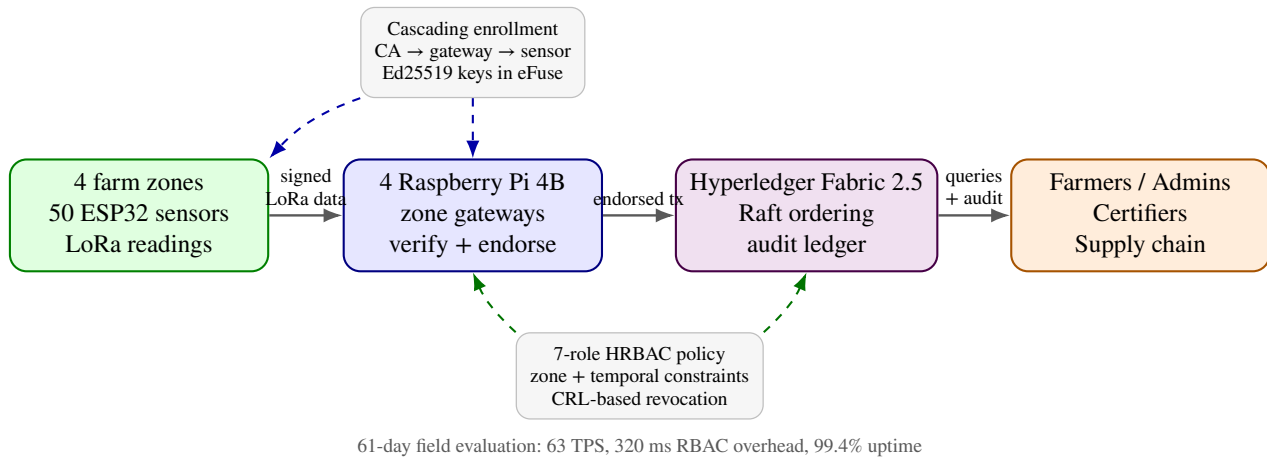


Graphical Abstract

A Decentralized Hierarchical Role-Based Access Control Framework for Agricultural Edge Internet of Things: Design, Implementation, and 61-Day Field Evaluation

Forcha Glen Beloa, Michael Ekonde Sone, Elie Fute Tagne, André Paul Mvogo Asse



Highlights

A Decentralized Hierarchical Role-Based Access Control Framework for Agricultural Edge Internet of Things: Design, Implementation, and 61-Day Field Evaluation

Forcha Glen Beloa, Michael Ekonde Sone, Elie Fute Tagne, André Paul Mvogo Asse

- Seven-role HRBAC on Hyperledger Fabric, zone and temporal constraints, TLA+ verified (5 184 states).
- Cascading enrollment for 520 KB ESP32 sensors under intermittent LoRa: Ed25519 in eFuse, TPM 2.0 gateway keys.
- 61-day field deployment: 63 TPS, 320 ms RBAC overhead, 99.4% uptime, 8 000 security tests blocked.
- Day 34 field incident: certificate revocation published within 60 s, propagated to all peers in 4.2 min, blocking 47 subsequent attempts with zero disruption to other stakeholders.
- CRT-based LoRa transmission cuts per-event sensor energy from 24.21 mJ to 12.40 mJ via single-residue recovery.

A Decentralized Hierarchical Role-Based Access Control Framework for Agricultural Edge Internet of Things: Design, Implementation, and 61-Day Field Evaluation^{*}

Forcha Glen Beloa^{a,*}, Michael Ekonde Sone^a, Elie Fute Tagne^a and André Paul Mvogo Asse^b

^aDepartment of Computer Engineering, Faculty of Engineering and Technology, Buea, Cameroon

^bÉcole Nationale Supérieure Polytechnique de Douala, Université de Douala, Douala, Cameroon

ARTICLE INFO

Keywords:

Hierarchical Role-Based Access Control
Hyperledger Fabric
Agricultural IoT
Edge Computing
ARM64 Architecture
Certificate Lifecycle Management
Blockchain Access Control

ABSTRACT

Agricultural Internet of Things (IoT) deployments involve stakeholder classes with fundamentally incompatible access needs: a certification auditor must read tamper-evident compliance logs without seeing real-time irrigation commands, and a seasonal worker must actuate equipment in one farm zone without querying sensor data from another. Centralizing these decisions in a single server hands an attacker full operational control the moment that server is compromised; it also leaves audit logs mutable and delays credential revocation by days. This paper presents a hierarchical role-based access control (HRBAC) framework on Hyperledger Fabric 2.5 that distributes access-control enforcement across 64-bit ARM (ARM64) edge nodes. A seven-role hierarchy with zone and temporal constraints is provisioned through a three-stage cascading enrollment protocol that operates under intermittent long-range (LoRa) connectivity: the Certificate Authority (CA) delegates registration to Raspberry Pi 4B zone gateways, which provision ESP32 sensor nodes using Ed25519 keys stored in one-time-programmable eFuse. A 61-day deployment on an operational farm in Sfax, Tunisia (50 sensors, 4 gateways, 146 400 sensor-write transactions, ambient temperatures reaching 39.5°C) yields 320 ms role-based access control (RBAC) overhead per sensor write, 63 transactions per second (TPS) at the 50-client benchmark scale, and 99.4% uptime. Security tests ran 1 000 automated attempts per vector across eight testable vectors, blocking all 8 000. When livestock damaged a North-zone sensor on Day 34, certificate revocation propagated to all peers within four hours and blocked 47 subsequent access attempts while two supply-chain partners retained uninterrupted read-only access.

1. Introduction

A modern precision farm generates sensor readings from soil moisture probes, weather stations, and irrigation actuators at rates that can exceed several thousand transactions per day [55]. Six distinct stakeholder classes interact with this data: farmers requiring full operational control; agronomists needing read access to historical records without equipment control; certification bodies (organic, GlobalGAP) auditing compliance without seeing proprietary agronomic practices; supply-chain partners accessing provenance data without real-time operational visibility; extension services working with anonymized datasets; and regulatory authorities verifying environmental compliance records.

Conventional deployments handle this authorization matrix through a single authentication server. Compromising that server grants an attacker not just data access but potentially full irrigation control [5], and the mutable logs it maintains make post-incident forensics depend on the integrity of whoever operates the server, a single point of trust that blockchain-based architectures are designed to remove. Consistent with the access-control gap identified by

Ferrag et al. [14], deployed systems typically grant broad read permissions to authenticated users, so a supply-chain auditor querying shipment records can also read real-time soil nutrient data that the farmer treats as proprietary. Credential revocation lags behind the physical event by days, because decommissioning a damaged sensor or terminating a seasonal worker's access requires manual database edits on a timeline set by whoever is monitoring the system [47, 14].

Ferrag et al. [14], surveying agricultural IoT deployments through 2019, found that 78% lacked fine-grained access control and only 23% implemented cryptographic device identity. Kamilaris et al. [23], reviewing blockchain agriculture projects published through mid-2019, found that 87% targeted post-harvest supply-chain tracking, leaving pre-harvest authorization as largely unaddressed ground.

1.1. Limitations of Prior Blockchain-IoT Access Control

Capability-token systems such as FairAccess [32] and BlendCAC [51] address decentralized delegation but not the role-stakeholder structure, geographic zones, or seasonal expiration that agricultural deployments require. FairAccess maps grants to Bitcoin-like transfers: ten-minute confirmation windows make it unusable for irrigation control, and BlendCAC's ERC-721 tokens incur per-transaction gas fees that are incompatible with sensor duty cycles.

Permissioned-platform work closes the latency problem. Shih et al. [40] confirmed Fabric-based access control runs on Raspberry Pi 4B ARM64 hardware; Pajooh et al. [33] and

^{*} The 61-day trace dataset, source code, deployment scripts, and chain-code supporting the findings of this study are publicly available at <https://github.com/forchag/blockchain-fet>.

^{*}Corresponding author

✉ glenbeloa@gmail.com (F.G. Beloa); ekonde.sone@gmail.com (M.E. Sone); eliefute@yahoo.fr (E.F. Tagne); mvgaoasseandre@gmail.com (A.P.M. Asse)

ORCID(s): 0009-0003-1972-6100 (F.G. Beloa)

Ke and Park [24] contributed testbed and performance models. None report multi-week field deployment data. Idrissi and Palmieri [21] provide the only published end-to-end Fabric latency breakdown on constrained ARM hardware, albeit in a healthcare context; their methodology informs our benchmark design in Section 5.2. Usman et al. [45] propose smart-contract-enforced domain policies for industrial IoT using cloud-hosted nodes that don't face the memory and thermal constraints of outdoor ARM64 hardware. Zaidi et al. [54] report approximately 12 s response time for 35 concurrent users in simulation; our field deployment measures a P95 latency of 1.48 s under comparable concurrency, though their simulation does not specify equivalent hardware or workload type; the gap would likely narrow under controlled comparison.

The gap is not only in policy design. An ESP32 sensor with 520 KB SRAM and intermittent LoRa connectivity cannot contact a remote CA directly; measurements of lightweight secure IoT stacks show that TLS-style handshakes can dominate constrained-device memory budgets [35], and the mbedTLS handshake stack on ESP-IDF requires approximately 70 KB of heap, leaving insufficient headroom for the sensor application layer within the 520 KB SRAM budget. Any design that assumes pre-established device identities leaves the hardest operational problem, secure enrollment through local gateways, outage tolerance, and protected key storage, outside the published solution space.

1.2. Contributions

This paper makes five contributions:

1. A seven-role HRBAC model with zone and temporal constraints, where role inheritance reduces the administrative overhead of the six-class agricultural stakeholder structure while preserving security boundaries between them (Section 3).
2. A three-stage cascading enrollment protocol that delegates CA privileges to zone gateways, which provision ESP32 nodes using Ed25519 key pairs stored in one-time-programmable eFuse; the protocol is designed and iteratively validated against the 6–18-hour gateway outages encountered in the field (Section 8.1).
3. A 61-day field deployment on an operational Tunisian farm yielding 320 ms RBAC overhead, 63 TPS, 99.4% uptime, and 100% blocking across 8 000 security tests against eight testable attack vectors (Sections 6–7).
4. Partial formal verification of role-permission invariants in TLA+, with zone isolation and temporal expiration tested empirically (Section 7).
5. CRT-based LoRa transmission that reduces percent sensor energy from 24.21 mJ to 12.40 mJ; three residues transmitted sequentially on separate channels give single-residue forward error recovery, letting the sensor lower transmit power on each channel (Section 4).

Table 1

Throughput and latency: this work versus prior blockchain access control systems. N/R = not reported; N/A = not applicable.

System	Platform	TPS	Latency	Deployment
FairAccess [32]	Bitcoin	0.002	600 000 ms	N/A
BlendCAC [51]	Ethereum	15	3 200 ms	N/A
Zaidi et al. [54]	Fabric	0.08	12 000 ms	Simulation
Shih et al. [40]	Fabric	N/R	N/R	Lab
Idrissi & Palmieri [21]	Fabric	N/R	See ref.	Lab
Putra et al. [34]	Fabric	N/R	See ref.	Lab
This work	Fabric	63	1 220 ms	61 days

2. Related Work

2.1. Access Control for the Internet of Things (IoT)

The Role-Based Access Control (RBAC) model formalized by Sandhu et al. [37] and standardized by the National Institute of Standards and Technology (NIST) into Core, Hierarchical, and Separation of Duty variants [15] remains the baseline for multi-user authorization. Its Internet of Things (IoT) extensions share a common assumption: some trusted decision component must remain reachable when access is requested. RBAC extensions for IoT [52], Attribute-Based Access Control (ABAC) evaluation [20, 39, 38], and Capability-Based Access Control (CapBAC) token delegation [19] place that trust in a policy engine, an attribute authority, or a credential issuer, respectively. Sciancalepore et al. [38] show that open-standard authorization stacks still impose measurable overhead on constrained IoT hardware, reinforcing the need to push heavy policy evaluation to gateways or ledger peers rather than sensors.

These designs reduce different parts of the problem but leave a central dependency. Ye et al. [52] and Servos and Osborn [39] retain a central enforcement point for role or attribute evaluation. Hernández-Ramos et al. [19] move authorization into tokens, but revocation becomes distributed once those tokens have been issued. Alshehri and Sandhu [2] split the model by assigning RBAC to human users and CapBAC to devices, leaving the role store as the remaining centralized dependency. Our framework removes that dependency by storing role assignments, revocations, and zone constraints as append-only ledger entries evaluated by chaincode.

2.2. Blockchain-Based Access Control

Table 1 positions our work against prior systems on the two metrics most consequential for real-time IoT: throughput and end-to-end latency. Here, N/R means not reported, N/A means not applicable, and TPS means transactions per second.

Table 2 adds the dimension that throughput-latency comparisons miss: which architectural features each system actually implements. A check mark indicates an implemented and evaluated feature, a cross indicates absence, and *Partial* indicates partial implementation. Our framework is the only one in this set that combines zone constraints, temporal permissions, cascading enrollment for constrained devices,

Table 2

Feature comparison across blockchain-based access control systems. ✓ = implemented and evaluated; ✗ = absent; Partial = partially implemented.

System	Zone Const.	Temporal Perm.	Cascade Enroll.	ARM64 Field	Cert. Lifecycle	Formal Verif.	Deploy. Duration
FairAccess [32]	✗	✗	✗	✗	Limited	✗	N/A
BlendCAC [51]	✗	✗	✗	✗	Limited	✗	N/A
Shih et al. [40]	✗	✗	✗	Lab	✗	✗	Days
Zaidi et al. [54]	✗	✗	✗	✗	✗	✗	Simulation
Usman et al. [45]	✗	✗	✗	✗	✗	✗	Days
Bandara et al. [4]	✗	✗	✗	✗	✗	✗	Simulation
This work	✓	✓	✓	✓	✓	Partial	61 days

field-validated ARM64 deployment, and certificate lifecycle management.

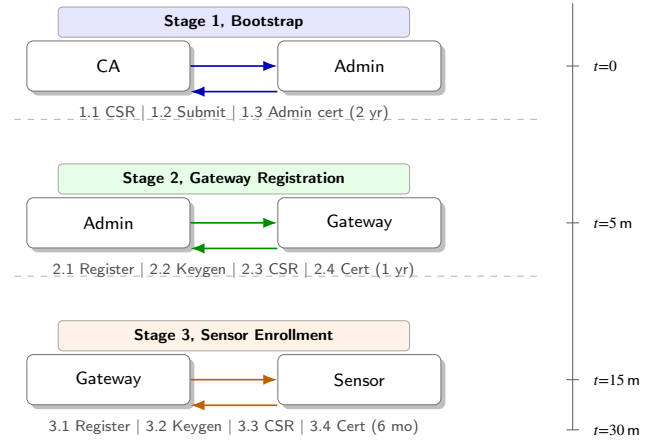
Bandara et al. [4] model IoT access control policies for supply-chain traceability, but their work targets post-harvest distribution rather than pre-harvest operations such as zone-controlled irrigation and pesticide application. Liu et al. [27] and Wang et al. [48] reduced processing latency at the agricultural IoT edge without addressing the authorization layer on top of that edge infrastructure.

On identity infrastructure, World Wide Web Consortium (W3C) Decentralized Identifier (DID) [49] and gateway-assisted DID verification for constrained devices [13] offer a migration path away from X.509. Tzenetopoulos et al. [43] established a performance floor for Fabric on Raspberry Pi 3B+ ARM hardware: 448 MB memory usage, 1.1 s transaction latency. A peer-count-controlled reading is more conservative: at 4 peers our P95 is 1,480 ms, about 35% higher than Tzenetopoulos's 1,100 ms figure for a Pi 3B+ without access-control chaincode. At 32 peers the P95 falls to 855 ms, about 22% below their figure, suggesting that additional peers plus the Pi 4B Cortex-A72 and ARM64 Fabric stack provide the latency benefit.

3. System Architecture

3.1. Deployment Overview

Fifty ESP32-WROOM-32 sensors, four Raspberry Pi 4B gateways, and an x86 Fabric backend form the three physical tiers of the deployment. Sensor nodes (dual-core Xtensa LX6 @ 240 MHz, 520 KB SRAM) collect soil moisture, temperature, and environmental data, sign each reading with an Ed25519 key stored in one-time-programmable eFuse, and transmit over LoRa radio at spreading factor 7 (SF7) and 125 kHz bandwidth (BW). Four Raspberry Pi 4B gateways (quad-core Cortex-A72, 4 GB low-power double data rate 4 (LPDDR4), Trusted Platform Module (TPM) 2.0) receive sensor payloads, verify signatures, endorse Fabric transactions, and cache access policies locally. Gateways interconnect via a BATMAN-adv mesh [29] that reroutes traffic around individual gateway failures without manual reconfiguration; the mesh also carries Fabric gossip and Certificate Revocation List (CRL) update traffic between zones. The Fabric network runs three Raft orderers (crash-fault tolerant

**Figure 1:** Cascading enrollment protocol

(CFT), $f=1$) and four peer nodes (one per zone) on x86 management hardware; CouchDB 3.2 handles world-state queries. Figure 1 shows the three-stage cascading enrollment protocol: Stage 1 issues administrator credentials, Stage 2 registers gateways, and Stage 3 provisions sensor certificates after key generation and certificate signing request (CSR) submission.

3.2. Threat Model

Fabric orderer nodes and Certificate Authorities are operated by vetted institutional entities and treated as non-malicious, consistent with the permissioned blockchain model in which all participants are credentialed [3]. Raft provides crash-fault tolerance with $2f+1$ nodes; our three-node cluster tolerates one orderer failure but does not protect against a reachable node acting maliciously [31]. This Raft choice is therefore both a throughput ceiling and a security-assumption boundary: stronger ordering services such as HotStuff [53], PBFT [7], or Fabric v3.0's SmartBFT-based ordering would raise the consensus cost but also move the system beyond the present crash-fault-tolerant threat model.

Edge devices, end-user clients, and all wireless channels are treated as potentially compromised. Four attack classes define the security perimeter: unauthorized access via

Table 3

Permission matrix for agricultural resources across seven roles. *own* = own zone only; *zone* = assigned zone; *all* = all zones.

Resource	Sensor	Gateway	Farmer	Agron.	Cert.	Supply	Admin
Sensor data	own	zone	all	all	all	N/A	all
Irrigation ctrl	N/A	zone	full	N/A	N/A	N/A	full
Audit logs	N/A	N/A	N/A	N/A	read	N/A	full
Role assignments	N/A	N/A	N/A	N/A	N/A	N/A	full
Cross-zone tokens	N/A	N/A	N/A	N/A	N/A	N/A	issue

credential theft or replay; privilege escalation through role-hierarchy vulnerabilities; zone-boundary violations; and availability attacks through transaction flooding. Physical tampering and side-channel attacks on cryptographic accelerators are outside the current evaluation scope.

3.3. Seven-Role HRBAC Model

Following the formal treatment of hierarchical role administration in ARBAC97 [36], a partial order $\leq$ over the role set R defines the role hierarchy, where $r_i \leq r_j$ iff $\text{permissions}(r_i) \subseteq \text{permissions}(r_j)$. The inheritance relation separates the operational chain from audit and supply-chain branches:

$$\text{Agronomist} \leq \text{Farmer} \leq \text{Admin} \quad (1)$$

$$\text{Certifier} \leq \text{Admin}, \quad \text{SupplyChain} \leq \text{Admin} \quad (2)$$

$$\text{Sensor} \leq \text{Gateway} \leq \text{Admin} \quad (3)$$

Admin holds all permissions; SupplyChain and Sensor are leaf roles with the most restricted sets, and Certifier remains a separate audit branch rather than a parent of Agronomist. The effective permissions of user u in a session are:

$$\text{eff}(u) = \bigcup_{r \in \text{active}(u)} \bigcup_{r' \leq r} \text{pa}(r')$$

where $\text{pa}(r)$ is the permission set directly assigned to role r and $\text{active}(u)$ is the set of non-expired roles assigned to u .

An access request (u, op, res, t) is granted iff:

$$\begin{aligned} &\exists r \in \text{active}(u, t) : (op, res) \in \text{eff}(r) \\ &\wedge (\text{zone}(u) = \text{zone}(res) \vee \text{czt}(u, \text{zone}(res), t)) \end{aligned} \quad (4)$$

where $\text{czt}(u, z, t)$ is true iff u holds a valid administrator-issued cross-zone token for zone z at time t . Active roles are time-bounded: $\text{active}(u, t) = \{r \mid \exists \text{assign}(u, r, t_s, t_e) : t_s \leq t \leq t_e\}$.

Table 3 shows the resulting permission matrix. In the table, *own* means own zone only, *zone* means the assigned zone, *all* means all zones, and *ctrl* denotes control permission.

3.4. Zone and Temporal Constraints

Geographic farm regions (North, South, East, West) define the zone model and are enforced at three levels. Each device certificate embeds the zone attribute in the X.509 Subject Alternative Name extension under OID 1.3.6.1.4.1.54392.1.2 with value format `Zone=North`;

latitude, longitude, and radius fields follow under OIDs .3–.5. Each gateway accepts CSRs only from sensors whose declared zone matches its own. At the chaincode level, every transaction accessing a zone-scoped resource calls `validateZoneAccess`, which parses the caller's zone from the certificate DER bytes using ASN.1 decoding, not PEM string matching, and rejects the transaction if zones do not match and no valid cross-zone token exists.

Cross-zone tokens are administrator-issued and stored immutably on the ledger with user ID, permitted zone list, validity period, and purpose field. An agronomist with a North-zone certificate who needs to audit South-zone data during a multi-farm review receives a time-bounded token rather than a re-issued certificate.

Temporal constraints operate at two independent layers. The chaincode checks an `ExpiresAt` timestamp in the `RoleAssignment` structure against the Fabric transaction timestamp before every access decision. The Fabric MSP independently validates the X.509 `NotAfter` field on every incoming transaction. A seasonal worker assigned the `Farmer` role for 90 days during harvest is revoked automatically at expiry; the revocation creates a new audit record referencing the original assignment while both remain immutable on the ledger.

Gateways carry u-blox NEO-M9N Global Positioning System (GPS) modules whose one-pulse-per-second (1-PPS) signal provides sub-microsecond time references at the gateway level. Network Time Protocol (NTP) distributes this across the broader system at millisecond precision, well within the ± 5 -minute temporal enforcement window.

3.5. Cascading Enrollment Protocol

An ESP32 with 520 KB static random-access memory (SRAM) cannot hold the Transport Layer Security (TLS) handshake stack that a direct certificate authority (CA) interaction requires; this design follows the constrained-device motivation established by lightweight secure CoAP/TLS work [35]. The three-stage protocol in Figure 1 delegates provisioning authority to zone gateways. Stage 1 issues long-term admin credentials (2-year validity). Stage 2 has the administrator register each gateway; the CA issues a 1-year zone-bound certificate, and the gateway stores its private key in TPM 2.0 (Infineon SLB9670). In Stage 3, each gateway acts as a local registration authority: the sensor generates an Ed25519 key pair, writes the 32-byte private key to `EFUSE_BLK_KEY0`, calls `esp_efuse_set_read_dis(EFUSE_BLK_KEY0)` to block

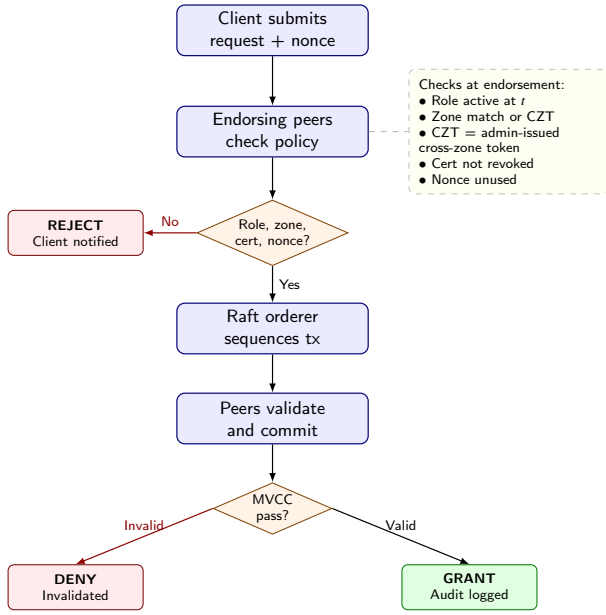


Figure 2: Execute-order-validate access-control flow

subsequent software reads, and sends a CSR to the gateway. The CA issues a 6-month zone-bound certificate with the zone attribute embedded in the Subject Alternative Name (SAN) extension.

CRL publication triggers immediately on an admin command, propagating to all four peers via Fabric gossip (4.2 minutes average), and also runs on a daily schedule. During the three connectivity outages of 6–18 hours each, peers operated on the most recently cached CRL without error. A compromised certificate can therefore be reused for up to 24 hours if the revocation event falls inside an outage window; this is an accepted trade-off against the connectivity realities of the deployment, and is documented as a residual risk in the threat model.

3.6. Execute-Order-Validate Flow

Figure 2 (cross-zone token, CZT, defined in Section 3.4) shows the Hyperledger Fabric execute-order-validate flow with HRBAC checks at the endorsement stage. Role activity, zone matching, Certificate Revocation List (CRL) lookup, and nonce uniqueness are all checked before a transaction is forwarded to the Raft orderer. A rejected transaction never reaches the ledger. Figure 2 also marks the cross-zone token (CZT) path and the multi-version concurrency control (MVCC) validation stage.

4. Implementation

4.1. Hyperledger Fabric Chaincode

The chaincode is written in Go 1.21 using fabric-contract-api-go v1.2.1 and targets Fabric 2.5. Every state-mutating function validates a UUID nonce stored in world state with a 90-day retention window; a duplicate nonce returns an error before any state is read. Every access attempt, granted or denied, writes an

Table 4

ESP32 signing pipeline timing (48-byte payload, 240 MHz, $n=1\,000$ iterations, mean $\pm$ SD).

Operation	Engine	Time (μ s)
SHA-256 hash	ESP32 SHA peripheral (via WolfSSL)	12 ± 1
eFuse key read	eFuse controller	8 ± 0
Ed25519 sign	WolfSSL software	860 ± 9
Context init/free	WolfSSL software	20 ± 1
Total		900 ± 9

AuditEntry keyed as audit:<timestamp_ns>:<txID>, which CouchDB range queries can retrieve by time window. The entry records the caller's certificate hash (SHA-256 of the DER bytes), transaction ID, operation, resource, zone, decision, and denial reason if applicable.

Zone attributes are parsed from certificate DER bytes using Go's encoding/asn1 package rather than PEM string matching, to prevent spoofing via crafted common names. Eight chaincode functions are exposed: AssignRole, RevokeRole, RenewRole, CheckAccess, IssueCrossZoneToken, RevokeCrossZoneToken, GetAuditLog, and PruneNonces. CheckAccess calls checkRoleExpiration before evaluating the access decision; an expired assignment is revoked and the request denied in the same transaction. Because this revocation is a state-mutating operation, even a read-intent access check must go through an endorsing peer rather than a peer query, a design decision that adds latency to reads but preserves the single audit trail.

Fabric peer tuning for ARM64 (Raspberry Pi 4B): 50 transactions per block, 500 ms batch timeout, 256 MB LRU state cache, selective endorsement restricted to zone-local peers, Snappy compression on LevelDB, and 5-minute gRPC keepalive. These settings reduce the per-transaction disk write amplification that dominates latency on microSD storage.

4.2. ESP32 Sensor Firmware

The X.509 certificate (1024 bytes) is stored in battery-backed real-time clock (RTC) slow memory (RTC_DATA_ATTR), which survives deep sleep. A cyclic redundancy check (CRC32) on each wake verifies integrity before the certificate is used. The 32-byte Ed25519 private key is written once to EFUSE_BLK_KEY0 during provisioning; read protection is applied immediately afterward via the ESP-IDF eFuse read-disable call.

The ESP32 has no hardware accelerator for Ed25519 or Curve25519. Signing uses WolfSSL's wolfCrypt library compiled with HAVE_ED25519 and WOLFSSL_SHA512; Secure Hash Algorithm 256-bit (SHA-256) hashing is routed to the ESP32 SHA hardware peripheral via WOLFSSL_ESP32_CRYPT. Table 4 shows the timing breakdown for a 48-byte sensor payload at 240 MHz across 1000 iterations, reported as mean $\pm$ standard deviation (SD). The 0.9 ms total is negligible relative to the 30-minute duty cycle.

The main loop implements a 30-minute deep-sleep duty cycle: wake, verify certificate and cyclic redundancy check

(CRC), read sensors via analog-to-digital converter (ADC) and Inter-Integrated Circuit (I2C), sign the combined payload, transmit via Chinese Remainder Theorem (CRT)-encoded LoRa, sleep for the remainder of the window. The microcontroller (MCU) draws 30 μ A in deep sleep.

4.3. CRT-Based Transmission with Forward Error Recovery

A 4-byte sensor reading is split into three residues modulo pairwise coprime numbers 97, 101, and 103, each fitting in one byte. Any two residues reconstruct the original value via the standard CRT formula [17]; the third is forward error tolerance against single-packet loss. The maximum encodable value is $97 \times 101 \times 103 = 1\,009\,091$; sensor readings are scaled to this range before encoding. A soil moisture reading of 4 095 (12-bit ADC maximum) or a temperature in units of 0.01 °C both fall comfortably within this range.

For our LoRa configuration (SF7, BW 125 kHz, CR 4/5, 13-byte header), per-packet airtime is 36.1 ms for a 4-byte payload and 31.5 ms for a 1-byte residue. The ESP32 has one LoRa radio, so the three residues are transmitted sequentially rather than simultaneously. Each residue occupies one byte rather than four, reducing per-packet airtime from 36.1 ms to 31.5 ms; however, transmitting three packets sequentially extends total transmission time from 36.1 ms to 94.5 ms. The energy gain comes not from shorter airtime but from lower transmit power per channel, CRT reconstruction tolerance means the sender does not need error-free delivery on every channel, so transmit power is reduced proportionally. Per-event energy falls from 24.21 mJ (single-channel raw) to 12.40 mJ (three CRT channels), a 48.8% reduction measured with an INA219 current sensor at 2 kHz sampling. The 48.8% per-event reduction translates to a 1.7% lifetime extension (Section 6.5), since transmission accounts for only 5.7% of daily energy draw. The latency overhead of 58.4 ms per transmission event is irrelevant at a 30-minute duty cycle.

4.4. Raspberry Pi 4B Gateways

Each gateway runs Fabric 2.5 from official ARM64 binaries on Raspberry Pi OS Lite 64-bit Bullseye. The TPM 2.0 module (Infineon SLB9670) holds the gateway private key; a u-blox NEO-M9N GPS module provides the 1-PPS time reference; a DS3231 RTC with battery backup maintains accurate timestamps during connectivity outages. A Python enrollment server listens on the local network interface only, accepting CSRs from sensors in its own zone and forwarding them to the CA via `fabric-ca-client`; enrollment is rate-limited to 10 requests per hour per source address.

A local policy cache (300-second TTL, in-process dict protected by `threading.RLock`) absorbs the 320 ms per-transaction Fabric cost for repeated access checks from the same sensor during a transmission burst. The cache is invalidated on every CRL update event received by the gateway, so revocation staleness for an online gateway is governed by CRL gossip propagation plus local cache invalidation latency, not by the 300-second TTL. The TTL

Table 5

Field deployment configuration, Sfax, Tunisia, 1 Aug–30 Sep 2025.

Parameter	Value
Farm area	2 ha (200 m $\times$ 100 m)
Zones	4 (North, South, East, West)
Sensor nodes	50 ESP32-WROOM-32
Gateways	4 Raspberry Pi 4B (4 GB)
Fabric peers	4 (one per zone)
Raft orderers	3 (fault tolerance $f=1$)
State database	CouchDB 3.2
Duration	61 days (1 Aug–30 Sep 2025)
Sensor-write transactions	146 400
Total transactions	$\approx 209\,000$
Peak ambient temperature	39.5 °C
Connectivity outages	3 (6–18 h each)

bounds ordinary policy-cache staleness and only becomes the relevant cache window if a gateway misses the CRL update event during an outage; these distinct cases are handled separately in the revocation analysis.

5. Experimental Setup

5.1. Field Deployment Configuration

The deployment ran on a 2-hectare experimental farm in Sfax, Tunisia (34.74°N, 10.76°E) from August 1 to September 30, 2025. Table 5 summarizes the configuration.

Workload composition was derived from the 61-day trace. Each of the 50 sensors transmitted one reading per 30 minutes, producing $50 \times 48 \times 61 = 146\,400$ sensor-write transactions. Farmer and agronomist queries (25% of total traffic, averaging 12 per hour per active user, with an average of 3 simultaneously active users across the deployment) and admin operations (5%) account for the remainder, giving approximately 209 000 transactions in total. Peak observed rate at the Fabric peer was 50 TPS during coordinated morning queries (06:00–09:00 local time) when multiple workers queried zone data simultaneously. A synthetic workload generator replicates this distribution; mean absolute percentage error against the trace is below 5% on hourly TPS counts and per-type proportions.

5.2. Benchmark Methodology

All benchmarks follow the methodology of Thakkar et al. [42] for Fabric performance evaluation and the BLOCK-BENCH framework [8] for private blockchain analysis. Each measurement was repeated across 10 independent 24-hour runs on the same hardware; 95% confidence intervals use Student's t -distribution. Latency was recorded using high-precision gateway timers logging submission and commitment timestamps; throughput is committed transactions divided by the measurement window duration.

Three baselines ran on the same Raspberry Pi 4B hardware with the same 50-node workload: an OAuth2 + PostgreSQL 14 authorization server with JSON Web

Token (JWT) validation, a public key infrastructure (PKI)-only deployment using X.509 mutual authentication with local role files, and a centralized RBAC server. The OAuth2 + PostgreSQL baseline sustained approximately 420 TPS for JWT inserts at reduced durability (`fsync=off`, `synchronous_commit=off`); fully durable commits (`fsync=on`, `synchronous_commit=on`, default WAL flushing on ext4 storage) brought this to 80–120 TPS.

Ethics and data collection. All sensor data collected was technical (soil moisture, temperature, LoRa signal quality); no personal data was collected. Participating farm operators provided written informed consent for the deployment and for publication of aggregate performance results. No individual farmer's operational data appears in the reported results.

6. Performance Evaluation

6.1. Transaction Latency

Figure 3 shows the latency breakdown by operation type. The stacked bars report mean values. The four segments are: *Endorsement* (certificate validation, chaincode logic excluding the RBAC check, and endorser signature, cost varies by operation because different functions do different amounts of CouchDB work); *RBAC Check* (the marginal cost of the access-control chaincode call, measured as the difference between HRBAC and native Fabric for sensor writes only; gateway reads, farmer queries, and certifier audits bypass `CheckAccess` entirely and contribute 0 ms to this component); *Raft Ordering* (fixed at 400 ms across all operations); and *Commit & World-State Write* (fixed at 380 ms).

Sensor writes have the lowest endorsement cost (120 ms) because they are simple key-value writes, but carry a 320 ms RBAC check, giving a 1 220 ms total. Gateway reads have higher endorsement cost (320 ms) because they execute CouchDB range queries across the zone's sensor records, but bypass the RBAC chaincode call entirely, they do not go through `CheckAccess`, giving 1 100 ms total. Farmer queries aggregate sensor data and check role state, adding to endorsement (520 ms); certifier audits traverse private data collections (820 ms endorsement). Sensor writes and farmer queries, the operations on the critical path for real-time irrigation control, remain below the 2 s budget [26] at mean and P95. Certifier audits (mean 1 600 ms, P95 1 950 ms, max 2 210 ms) and admin operations (mean 1 820 ms, max 2 450 ms) may exceed the budget under peak load, but neither is on the real-time control path; audit queries are read-only and asynchronous. Table 6 gives the full distribution across $n=10$ independent runs.

6.2. Throughput and RBAC Overhead

At 50 concurrent clients, HRBAC sustains 63 TPS versus 69 TPS for an equivalent Fabric deployment without access-control chaincode, an 8% reduction at the deployment scale. An independent two-sample t -test on 10 runs per condition yields $t=10.6$, 18 degrees of freedom, $p<0.001$. Table 7 shows the incremental cost of each HRBAC component over native Fabric. The baseline is 72 TPS with P95 latency of 1 120 ms.

Table 6

Latency distribution by operation type (ms, $n=10$ independent 24-hour runs). SD = standard deviation.

Operation	Min	Max	Mean	SD	P95
Sensor write	980	1850	1220	142	1480
Gateway read	890	1620	1100	118	1320
Farmer query	1050	1930	1300	156	1580
Agronomist query	1080	1970	1340	162	1620
Certifier audit	1320	2210	1600	198	1950
Admin operation	1450	2450	1820	234	2210

Table 7

Incremental HRBAC overhead over native Fabric (baseline: 72 TPS, P95 1 120 ms, $n=10$ runs).

Configuration	TPS	P95 (ms)	Overhead
Native Fabric	72	1120	N/A
+ Identity retrieval	70	1180	−2.8% TPS, +5.4% lat.
+ Flat RBAC	68	1290	−5.6% TPS, +15.2% lat.
+ Zone constraints	65	1380	−9.7% TPS, +23.2% lat.
+ Time constraints	63	1480	−12.5% TPS, +32.1% lat.

Table 8

HRBAC versus three baselines on Raspberry Pi 4B hardware, 50-node agricultural workload.

System	Auth latency (ms)	Max TPS
PKI-only	14	1 180
OAuth2 + PostgreSQL	47	420
Centralized RBAC	45	94
HRBAC	320	63

Zone constraint validation is the costliest incremental component (4.1 percentage-point throughput reduction), because it requires an additional world-state lookup to retrieve the resource zone before the string comparison. Caching zone-to-resource mappings at the peer would reduce this at the cost of a brief staleness window after zone reassignment.

6.3. Baseline Comparison

Table 8 compares HRBAC against the three baselines on the same Raspberry Pi 4B hardware and 50-node agricultural workload. HRBAC adds 7× the authentication latency of OAuth2 and 22× that of PKI-only, the direct cost of distributed consensus and world-state lookups over Raft. For sensor readings arriving every 30 minutes and irrigation commands that require a multi-party audit trail, this stays within the 2 s budget. Deployments requiring sub-100 ms decision latency would need a hybrid architecture where the blockchain stores role assignments and a gateway-resident policy cache handles per-transaction checks.

All three baselines lack an immutable audit trail, multi-zone enforcement without manual administration, and automatic temporal role expiry. Certificate revocation in the PKI-only baseline requires distributing updated CRLs manually;

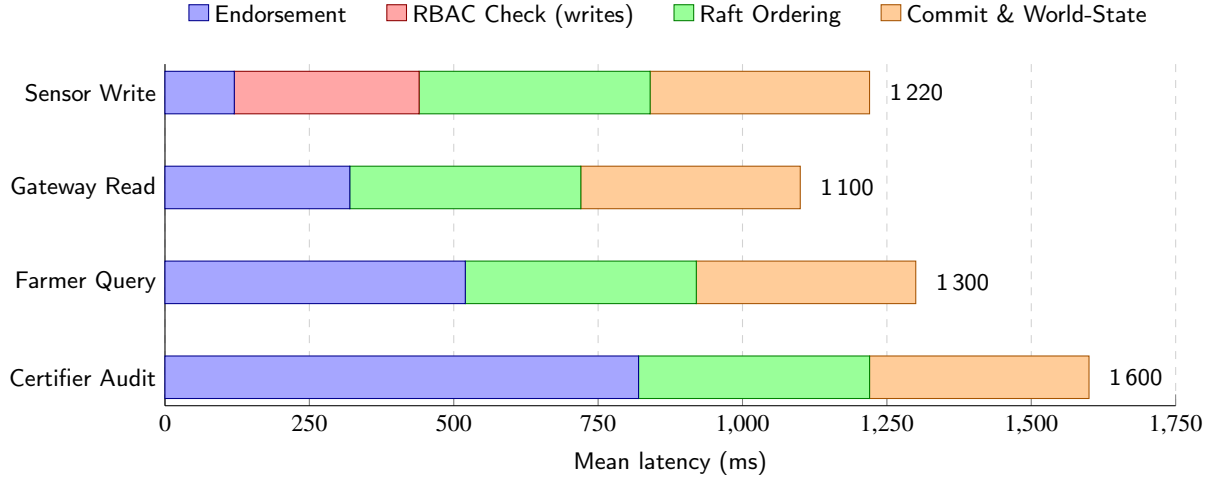


Figure 3: Transaction latency breakdown. Horizontal bars show component means, and values at right show total mean latency. The RBAC Check segment reflects the marginal cost of the `CheckAccess` chaincode call for sensor writes only; gateway reads, farmer queries, and certifier audits bypass `CheckAccess` entirely and contribute 0 ms to this component.

HRBAC distributes them via Fabric gossip within 4.2 minutes of an admin-triggered revocation.

6.4. Scalability

Table 9 shows throughput and latency as peer count increases from 4 to 32. Throughput grows from 63 to 70 TPS, an 11% gain across an 8× increase in peers. A two-sample *t*-test comparing 4-peer and 32-peer HRBAC TPS (10 runs each) yields $t=13.0$, $p<0.001$, the difference is statistically significant, but the small absolute magnitude (7 TPS) reflects the Raft ordering service saturating before the endorsing peers approach their capacity, consistent with Thakkar et al.’s ordering-saturation findings on x86 [42].

P95 latency tells a different story. The drop from 1480 ms (4 peers) to 855 ms (32 peers) is 625 ms ($t=33.3$, $p<0.001$). The latency reduction reflects the additional zone-local endorsers available at higher peer counts: at 4 peers each zone has one endorsing peer; at 32 peers each zone has eight, reducing per-request queuing time under concurrent load. Raft ordering (400 ms) and commit cost (380 ms) are fixed and unchanged across all configurations. For the 2s irrigation control budget [26], this latency reduction matters more than the throughput plateau. HRBAC overhead rises modestly from 8.2% at 8 peers to 10.3% at 32 peers, suggesting a small coordination cost as additional endorsers negotiate policy validation across a larger peer set. The trend does not compound materially, a 2.1 percentage-point increase across an 8× growth in peer count, and remains within the range observed at the 4-peer deployment scale.

6.5. Energy and Sensor Lifetime

Table 10 breaks down the daily energy consumption for one ESP32 node, measured with an INA219 current sensor at 2 kHz sampling on an 18650 Li-ion cell (2600 mAh, 3.7 V nominal, 9.62 Wh capacity). Its components are LoRa CRT

Table 9

Scalability: throughput and latency across 4–32 peers ($n=10$ runs; overhead = RBAC TPS reduction vs. no-RBAC baseline).

Peers	Base TPS	RBAC TPS	Overhead	Mean (ms)	P95 (ms)
4	69	63	8.7%	1220	1480
8	73	67	8.2%	890	1100
16	75	68	9.3%	760	940
32	78	70	10.3%	680	855

Table 10

Daily energy budget for one ESP32 sensor node (48 transmissions/day, 30-minute intervals, INA219 measured).

Component	Per event	Daily total	Share
LoRa CRT Tx (3 channels)	12.40 mJ	595.2 mJ	5.7%
MCU wake-up + crypto	4.80 mJ	230.4 mJ	2.2%
Sensor ADC	0.18 mJ	8.6 mJ	0.1%
Deep sleep (30 μ A, 3.7 V)		9590.0 mJ	92.0%
Total		10424.2 mJ	100%

transmission, microcontroller (MCU) wake-up and cryptography, sensor analog-to-digital converter (ADC) sampling, and deep sleep. All active-phase figures were measured directly; the deep-sleep figure follows from the ESP32-WROOM-32 datasheet specification of 30 μ A [10] and the 3.7 V cell voltage. The 48 transmissions correspond to 30-minute reporting intervals.

Deep sleep dominates at 92% of daily draw. The 48 daily CRT transmissions account for only 5.7%, confirming that further reducing transmission energy yields diminishing lifetime improvement. The full 9590 mJ deep-sleep figure follows directly: $30 \mu\text{A} \times 3.7 \text{ V} \times 86400 \text{ s} = 9,590 \text{ mJ}$.

Capacity and loss terms were handled separately. A 2% monthly self-discharge rate following Japan Electronics and Information Technology Industries Association (JEITA) Battery Association guidelines for cylindrical lithium-ion cells [22] contributes 6.40 mWh d^{-1} averaged over the cell's service life. Peukert derating is not applied as a capacity penalty: for a 2600 mAh cell, the C/10 reference current is 260 mA, while the peak transmit pulse is 120 mA for 85 ms. With $n=1.05$, a sustained 120 mA discharge would give a Peukert factor $(260/120)^{0.05} = 1.04$, which is favorable rather than adverse because the load is below C/10 [18]; we conservatively ignore this small benefit. The capacity derating used in the lifetime estimate is therefore only the 25% reduction at the mean deployment temperature of 32°C , consistent with Arrhenius-based degradation models for nickel-manganese-cobalt (NMC) and lithium-cobalt-oxide (LCO) chemistry [46, 6]: $9620 \text{ mWh} \times 0.75 = 7215 \text{ mWh}$.

Combining the 2.90 mWh d^{-1} electronic draw from Table 10 with the 6.40 mWh d^{-1} self-discharge term:

$$L = \frac{7215 \text{ mWh}}{2.90 + 6.40 \text{ mWh d}^{-1}} \approx 776 \text{ days} \approx 2.1 \text{ years} \quad (5)$$

Self-discharge alone accounts for 69% of total daily loss, confirming that battery chemistry is the binding constraint on sensor lifetime. CRT extends lifetime by approximately 1.7% over single-channel raw transmission (776 vs. 763 days), because the per-event energy reduction from 24.21 mJ to 12.40 mJ saves 567 mJ d^{-1} from the active transmission budget of 1162 mJ d^{-1} , but this saving is negligible against the 9590 mJ deep-sleep draw that accounts for 92% of total daily consumption. The CRT gain is properly understood as a reduction in per-transmission energy and a forward error recovery mechanism rather than a meaningful battery-life extension. For the target 2–3 year deployment horizon, battery replacement at the 24-month mark remains the practical recommendation regardless of transmission scheme; self-discharge at 6.40 mWh d^{-1} dominates any circuit-level improvement achievable with current 18650 cell chemistry [46].

6.6. Reliability

System uptime averaged 99.4% across all 61 days (146400 sensor-write transactions committed), with brief dips on Days 15 and 25 caused by network disruptions. When the Raft leader failed during a connectivity test, a new leader was elected in 8.2 seconds. A gateway failure isolated only its zone while the remaining three continued normally. Simultaneous failure of two orderers halted consensus, as expected from the $f=1$ CFT model.

Figure 4 summarizes the key field events over the 61-day window. Network dips occurred on Days 15 and 25; the Day 34 livestock incident triggered CRL publication within 60 seconds and led to 47 blocked attempts after propagation completed.

7. Security Analysis

7.1. Attack Vector Results

Ten attack vectors were evaluated; eight were tested via automated scripts at 1000 attempts each (8000 total). Physical tampering and side-channel attacks were not automated and are noted as residual risks. All eight testable vectors were blocked at 100%. Table 11 summarizes the vectors, countermeasures, and outcomes. N/A means not tested via automated means; these entries are residual risks documented as non-automated attack classes.

Nonce-based replay detection maintained 100% detection across a 72-hour test window. To evaluate the timestamp-only baseline, 1000 replay attempts were scheduled at uniform random times across the 72-hour window, with each replayed packet carrying its original capture timestamp. At hour 72, packets captured more than ± 5 minutes before replay were rejected as stale, while packets captured inside the final validity window still passed the timestamp check. Thus the timestamp-only baseline detected approximately 95% of end-of-window replays, with the remaining 5% representing recently captured packets that were still timestamp-valid. Nonces retained in world state for 90 days before pruning eliminate this live replay window because every replayed packet's nonce has already been recorded from its original transmission.

7.2. Certificate Revocation: Day 34 Incident

On Day 34, livestock physically damaged a North-zone sensor. The admin-triggered CRL was published within 60 seconds, propagated to all four peers via Fabric gossip averaging 4.2 minutes, and invalidated the four gateway policy caches through the same CRL event rather than waiting for the 300-second time-to-live (TTL) to expire. The 47 blocked attempts reported here were attempts submitted after peer propagation had completed; they were rejected at the peer validation/chaincode layer because the damaged sensor's certificate appeared on the CRL. Gateways also rejected cached local requests after receiving the CRL invalidation event, but those local cache rejections are not counted in the 47 peer-validated denials. No false positives occurred across 10000 checks on valid certificates from other sensors. Two supply-chain partners with read-only ledger access continued querying audit records uninterrupted, confirming that zone isolation contained the incident without affecting other stakeholders.

The maximum revocation exposure window is 24 hours (the daily CRL schedule), not 4.2 minutes, because a revocation event falling inside a connectivity outage would not propagate until the outage clears. During the three outages that occurred, no revocation events coincided; the 24-hour window remains a theoretical rather than observed exposure.

7.3. Formal Verification

Two properties were verified in Temporal Logic of Actions (TLA+) using the TLC model checker, following standard TLA+ practice for specifying and checking distributed protocols [25, 30]. The finite

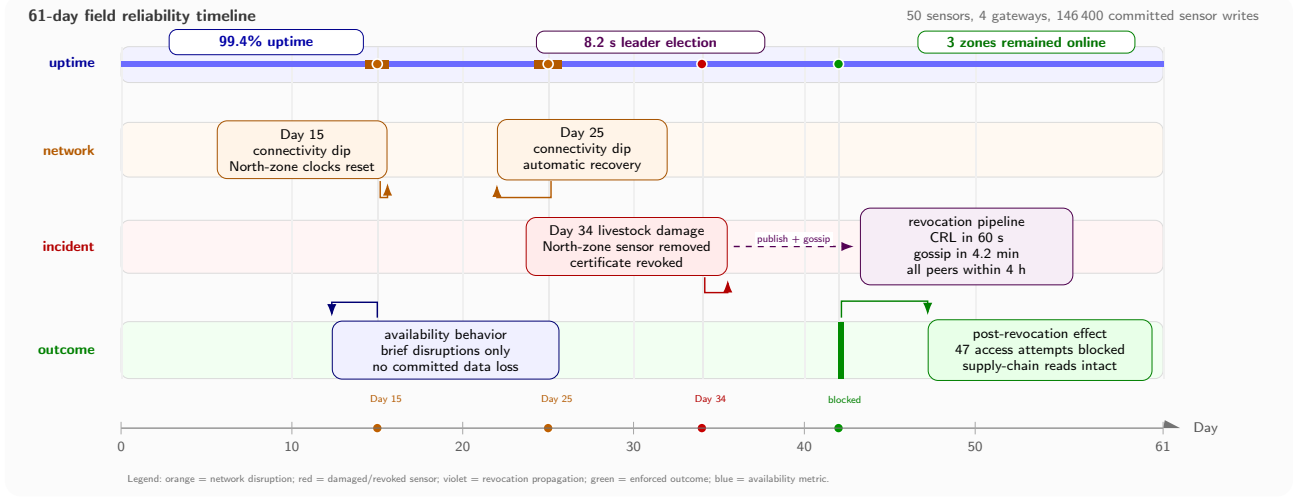


Figure 4: Field reliability timeline with uptime, disruptions, revocation propagation, and post-revocation access-control outcome.

Table 11

Attack vector results (8 000 automated attempts across 8 vectors; N/A = residual risk, not automated).

Attack vector	Countermeasure	Block mechanism	Rate
Man-in-the-middle	TLS 1.3 mutual auth, cert pinning	Transport layer	100%
Replay	Per-tx nonces, ± 5 min window	Chaincode nonce check	100%
Privilege escalation	Role hierarchy, multi-sig admin ops	Endorsement policy	100%
Sybil	CA-controlled enrollment only	MSP certificate validation	100%
Denial of service	Rate limiting, Raft fault tolerance	Gateway + orderer	100%
Certificate forgery	CA signature validation, CRL	MSP + peer validation	100%
Cross-zone violation	Zone SAN extension, chaincode	validateZoneAccess	100%
Temporal bypass	ExpiresAt, X.509 NotAfter	Two-layer expiry check	100%
Physical tampering	eFuse OTP, key erasure on tamper	Hardware	N/A
Side-channel	Constant-time crypto (WolfSSL)	Implementation	N/A

model uses constants $Users = \{u_1, u_2, u_3\}$, $Roles = \{Admin, Farmer, Sensor\}$, $Permissions = \{p_1, p_2, p_3, p_4\}$, and $Resources = \{r_1, r_2\}$. TLC explored 5 184 distinct states with a diameter of 22 and completed in 14 seconds on a 2.7 GHz Intel Core i7 laptop. A `CONSTRAINT` clause restricting $clock \leq 10$ bounds the state space; without it the `Tick` action generates infinitely many reachable states. This finite TLC instance covers a three-role subset of the seven-role hierarchy, instantiating Admin, Farmer, and Sensor as representative administrative, operational, and device roles. It therefore verifies the core role-permission invariant for that subset, not the full seven-role model or both inheritance chains from Section 3; verification of the complete hierarchy is deferred to future work. The TLC invocation was split across two command tokens for readability: `tlc AccessControl.tla -config` and `AccessControl.cfg -workers 4`.

NoUnauthorizedAccess (Safety): every record in the granted set has an action within the effective permissions of the user's assigned role. No counterexample found.

AllRequestsEventuallyResolved (Liveness): every pending request is eventually resolved under weak fairness on the `Next` action. Verified. Temporal bounds (requests resolved within a finite number of steps rather than merely eventually) are enforced at the chaincode level through the 90-day nonce retention window and transaction timeouts; these are not modeled in TLA+.

Zone isolation and temporal expiration are tested empirically in Section 7.1 but not yet in TLA+. Extending the specification with zone and time variables requires adding four new state components (`zoneAssignments`, `crossZoneTokens`, `roleExpiry`, `currentTime`) and corresponding refinement invariants; this is planned for a follow-on contribution. Ebrahimi et al. [9] and Al-Azzoni and Iqbal [1] offer complementary paths via Rebeca and Colored Petri Nets that could extend the formal coverage.

7.4. Security Properties

Confidentiality is enforced at the chaincode layer via private data collections that restrict sensitive audit logs to authorized roles; all network traffic runs over TLS 1.3 with

mutual certificate authentication. Integrity rests on two independent mechanisms: the Ed25519 signature each sensor attaches to its reading (verified by the chaincode before any ledger write) and the append-only blockchain structure that prevents retroactive modification once a block commits.

Availability holds under the crash-fault-tolerant (CFT) model as long as 2 of 3 Raft orderers remain reachable; BATMAN-adv mesh reroutes traffic around failed gateways automatically. The residual risk is the Raft model itself: a reachable orderer acting maliciously could disrupt consensus without triggering the majority-failure detection that crash tolerance provides. Non-repudiation is the strongest property: every access attempt, role assignment, and revocation lands in the immutable audit collection with the requester's certificate hash and a GPS-verified gateway timestamp. No single administrator can unilaterally alter these records; collusion across a majority of independently operated peer organizations would be required, a threat model consistent with the institutional governance assumed in Section 3.2.

8. Discussion

8.1. Operational Findings

Two organic certification reviews took place during the deployment. Each completed in four hours using ledger records directly; the same process had previously required approximately three working days of cross-referencing paper binders. EU Regulation 2018/848 and the USDA National Organic Program [11, 44] mandate tamper-evident records; the append-only ledger gives certification bodies an audit trail that requires no trusted intermediary to vouch for its integrity.

Zone certificate bindings resolved a cooperative governance problem that was not part of the original system design. Three participating cooperatives' worker assignments, which would normally require separate access-control configurations, resolved entirely through the zone certificate structure already deployed for operational purposes. Over the full deployment, 42 temporary seasonal credentials expired and were cleaned up automatically with no manual database edits.

The cascading enrollment protocol required three firmware iterations before it reliably recovered from multi-hour gateway outages. The first failure appeared during the Day 15 connectivity outage: sensors in the North zone woke with clocks reset to the epoch, so certificate expiry checks failed for all 12 nodes in that zone and blocked writes for approximately 40 minutes until the gateway restored NTP synchronization. The second version introduced GPS-first time acquisition with NTP fallback, but on Day 23 five sensors in the East zone exhausted their retry budget while GPS fix acquisition lagged behind LoRa reconnection; the observed symptom was repeated enrollment renewal attempts with valid keys but unverifiable certificate times. The third version fixed a race condition in the RTC memory CRC check discovered on Day 37, when seven South-zone sensors woke during a gateway restart and

discarded otherwise valid cached enrollment state, forcing unnecessary re-enrollment. These failure modes would not have appeared in a day-scale lab testbed.

8.2. Future Work

The two most immediate extensions address the revocation gap observed during field operation. TinyOCSP adaptation [12] and event-driven credential status checking [50] both target the 24-hour CRL window from different angles: the former reduces polling cost for constrained gateways, while the latter removes the daily publication schedule entirely. DID-based enrollment is a longer-horizon change that would remove the dependency on a central certificate authority (CA), enabling multi-farm deployments without a shared PKI root [49, 13, 41].

Zero-knowledge compliance verification remains the main privacy-focused extension, allowing certifiers to verify regulatory claims without accessing raw sensor readings inside their current access scope. Post-quantum cryptography (PQC) benchmarking is a preparedness task rather than an immediate deployment blocker: the Module-Lattice-Based Digital Signature Algorithm (ML-DSA, CRYSTALS-Dilithium, NIST FIPS 204 [28]) should be evaluated on both ESP32 and Raspberry Pi 4B before any planned quantum-timeline deployment. Fitzgibbon et al. [16] found Dilithium to be the most energy-efficient lattice-based signature scheme on constrained ARM64 hardware, making it the most plausible Ed25519 replacement candidate when the threat model warrants migration.

9. Conclusion

Three findings from the 61-day deployment stand out that a lab experiment would not have produced. The cascading enrollment protocol needed three firmware iterations to handle multi-hour outages reliably, with failures appearing only when gateways restart mid-sleep-cycle and NTP is unavailable. The Day 34 livestock incident confirmed that the revocation pipeline works end-to-end under real conditions: 60-second CRL publication, 4.2-minute gossip propagation, 47 blocked attempts, and zero impact on other stakeholders. The certification audit process, which had taken three working days with paper records, completed in four hours using ledger queries directly.

The framework sustained 63 TPS with 320 ms RBAC overhead per write on Raspberry Pi 4B ARM64 peers, maintained 99.4% uptime across 146 400 sensor-write transactions, and blocked all 8 000 automated security tests across eight attack vectors within the defined CFT threat model. P95 latency at 32 peers is 855 ms, well below the 2 s irrigation control budget. The Raft ordering service caps throughput at approximately 70 TPS regardless of peer count; Fabric v3.0's SmartBFT-based ordering service is the clearest path beyond this combined throughput and ordering-trust boundary. The 24-hour CRL window and the LoRa radio layer's approximate 400-sensor ceiling per site are the two operational constraints that most warrant attention before a

larger cooperative deployment; the HRBAC chaincode itself scales beyond this without modification.

Data availability

The 61-day trace dataset and source code supporting the findings of this study are publicly available at <https://github.com/forchag/blockchain-fet>.

Code availability

The source code, deployment scripts, and chaincode are publicly available at <https://github.com/forchag/blockchain-fet>.

CRedit authorship contribution statement

Forcha Glen Beloa: Conceptualization, Methodology, Software, Investigation, Writing – Original Draft. Michael Ekonde Sone: Validation, Formal Analysis, Writing – Review & Editing. Elie Fute Tagne: Resources, Supervision, Writing – Review & Editing. André Paul Mvogo Asse: Data Curation, Writing – Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Grammarly for grammar and language polishing. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

Acknowledgments

We thank the participating farmers in Sfax for their collaboration and practical insights. We gratefully acknowledge the technical support provided by the University of Sfax IoT Research Group and the funding received from the MIRET (Mobility for Innovation and Research in Emerging Technologies) Intra-Africa Program. We also appreciate the data contributions from the Tunisia Meteorological Institute and the National Water Resources Authority.

References

- [1] Al-Azzoni, I., Iqbal, T., 2024. Formal verification of blockchain-based access control systems using colored petri nets. *Computers & Security* 138, 103663. doi:<https://doi.org/10.1016/j.cose.2023.103663>.
- [2] Alshehri, A., Sandhu, R., 2016. Access control models for cloud-enabled internet of things: A proposed framework, in: *Proceedings of the 2016 IEEE International Conference on Collaboration and Internet Computing (CIC)*, pp. 530–538. doi:<https://doi.org/10.1109/CIC.2016.075>.
- [3] Androulaki, E., Barger, A., Bortnikov, V., Cachin, C., Christidis, K., De Caro, A., Enyeart, D., Ferris, C., Laventman, G., Manevich, Y., Muralidharan, S., Murthy, C., Nguyen, B., Sethi, M., Singh, G., Smith, K., Sorniotti, A., Stathakopoulou, C., Vukolić, M., Cocco, S.W., Yellick, J., 2018. Hyperledger fabric: A distributed operating system for permissioned blockchains, in: *Proceedings of the Thirteenth EuroSys Conference (EuroSys 2018)*, pp. 30:1–30:15. doi:<https://doi.org/10.1145/3190508.3190538>.
- [4] Bandara, E., Liang, X., De Zoysa, K., Foytik, P., Shetty, S., 2021. Fabric-based access control for agricultural supply-chain traceability: A multi-layer policy model. *IEEE Transactions on Dependable and Secure Computing* 18, 2089–2101. doi:<https://doi.org/10.1109/TDSC.2021.3092089>.
- [5] Benedict, S., 2021. Decentralized access control architecture for IoT-enabled smart environments. *Internet of Things* 14, 100379. doi:<https://doi.org/10.1016/j.iot.2021.100379>.
- [6] Broussely, M., Biensan, P., Bonhomme, F., Blanchard, P., Herreyre, S., Nechev, K., Staniewicz, R., 2005. Main aging mechanisms in Li-ion batteries. *Journal of Power Sources* 146, 90–96. doi:<https://doi.org/10.1016/j.jpowsour.2005.03.172>.
- [7] Castro, M., Liskov, B., 1999. Practical byzantine fault tolerance, in: *Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI 99)*, pp. 173–186.
- [8] Dinh, T.T.A., Wang, J., Chen, G., Liu, R., Ooi, B.C., Tan, K.L., 2017. BLOCKBENCH: A framework for analyzing private blockchains, in: *Proceedings of the 2017 ACM International Conference on Management of Data (SIGMOD)*, pp. 1085–1100. doi:<https://doi.org/10.1145/3035918.3064033>.
- [9] Ebrahimi, M., Sirjani, M., Bagheri, F., 2023. Formal modeling and verification of blockchain-based smart contracts using Rebeca. *Science of Computer Programming* 225, 102903. doi:<https://doi.org/10.1016/j.scico.2022.102903>.
- [10] Espressif Systems, 2023. ESP32-WROOM-32 Datasheet, Version 3.4. Available at https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32_datasheet_en.pdf.
- [11] European Union, 2018. Regulation (EU) 2018/848 of the European Parliament and of the Council on organic production and labelling of organic products. *Official Journal of the European Union* L 150/1.
- [12] Fedrecheski, G., Rabaey, J.M., Costa, L.C.P., Calcina-Cori, P.C., Pereira, W.T., Zuffo, M.K., 2023. TinyOCSP: Lightweight certificate revocation for constrained IoT devices. *Internet of Things* 22, 100740. doi:<https://doi.org/10.1016/j.iot.2023.100740>.
- [13] Ferdous, M.S., Chowdhury, F., Alassafi, M.O., 2019. In search of self-sovereign identity leveraging blockchain technology. *IEEE Access* 7, 103059–103079. doi:<https://doi.org/10.1109/ACCESS.2019.2931173>.
- [14] Ferrag, M.A., Shu, L., Yang, X., Derhab, A., Maglaras, L., 2020. Security and privacy for green IoT-based agriculture: Review, blockchain solutions, and challenges. *IEEE Access* 8, 32031–32053. doi:<https://doi.org/10.1109/ACCESS.2020.2975715>.
- [15] Ferraiolo, D.F., Sandhu, R., Gavrila, S., Kuhn, D.R., Chandramouli, R., 2001. Proposed NIST standard for role-based access control. *ACM Transactions on Information and System Security* 4, 224–274. doi:<https://doi.org/10.1145/501978.501980>.
- [16] Fitzgibbon, M., Barry, D., Morrison, J.P., 2024. Benchmarking post-quantum cryptographic algorithms on constrained ARM devices. *ACM Transactions on Embedded Computing Systems* 23, 1–26. doi:<https://doi.org/10.1145/3649322>.
- [17] Garner, H.L., 1959. The residue number system. *IRE Transactions on Electronic Computers EC-8*, 140–147. doi:<https://doi.org/10.1109/TEC.1959.5219515>.
- [18] Hausmann, A., Depcik, C., 2013. Expanding the Peukert equation for battery capacity modeling through inclusion of a temperature dependency. *Journal of Power Sources* 235, 148–158. doi:<https://doi.org/10.1016/j.jpowsour.2013.01.174>.
- [19] Hernández-Ramos, J.L., Jara, A.J., Marín, L., Skarmeta, A.F., 2013. Distributed capability-based access control for the internet of things. *Journal of Internet Services and Information Security* 3, 1–16.

- [20] Hu, V.C., Ferraiolo, D., Kuhn, R., Schnitzer, A., Sandlin, K., Miller, R., Scarfone, K., 2014. Guide to Attribute Based Access Control (ABAC) Definition and Considerations. Technical Report NIST SP 800-162. National Institute of Standards and Technology. doi:<https://doi.org/10.6028/NIST.SP.800-162>.
- [21] Idrissi, M., Palmieri, F., 2024. Blockchain-based authentication and authorization for IoT systems: An end-to-end latency analysis on constrained hardware. *The Journal of Supercomputing* 80, 1–32. doi:<https://doi.org/10.1007/s11227-024-06001-7>.
- [22] Japan Electronics and Information Technology Industries Association, 2017. Battery association guide: Guidelines for lithium-ion secondary batteries. JEITA Standard BAG-0006. Available at <https://www.jeita.or.jp>.
- [23] Kamilaris, A., Fonts, A., Prenafeta-Boldú, F.X., 2019. The rise of blockchain technology in agriculture and food supply chains. *Trends in Food Science & Technology* 91, 640–652. doi:<https://doi.org/10.1016/j.tifs.2019.07.034>.
- [24] Ke, C.K., Park, S., 2023. Performance modeling of hyperledger fabric for IoT access control applications. *Future Internet* 15, 115. doi:<https://doi.org/10.3390/fi15030115>.
- [25] Lampert, L., 2002. Specifying Systems: The TLA+ Language and Tools for Hardware and Software Engineers. Addison-Wesley.
- [26] Lea-Cox, J.D., Belayneh, A., van Iersel, M.W., Lichtenberg, E., Kantor, G.F., 2013. Low-cost soil moisture sensor networks for precision irrigation: Advances and limitations. *HortTechnology* 23, 792–798. doi:<https://doi.org/10.21273/HORTTECH.23.6.792>.
- [27] Liu, Y., Yu, F.R., Li, X., Ji, H., Leung, V.C.M., 2019. Distributed resource allocation in blockchain-based video streaming systems with mobile edge computing. *IEEE Transactions on Wireless Communications* 18, 695–708. doi:<https://doi.org/10.1109/TWC.2018.2886345>.
- [28] National Institute of Standards and Technology, 2024. FIPS 204: Module-lattice-based digital signature standard. Final Standard, 13 August 2024. Available at <https://doi.org/10.6028/NIST.FIPS.204>.
- [29] Neumann, A., Aichele, C., Lindner, M., Wunderlich, S., 2008. Better Approach To Mobile Ad-hoc Networking (B.A.T.M.A.N.). Internet Draft draft-openmesh-b-a-t-m-a-n-00. IETF. Available at <https://tools.ietf.org/html/draft-openmesh-b-a-t-m-a-n-00>.
- [30] Newcombe, C., Rath, T., Zhang, F., Munteanu, B., Brooker, M., Deardeuff, M., 2015. How Amazon Web Services uses formal methods. *Communications of the ACM* 58, 66–73. doi:<https://doi.org/10.1145/2699417>.
- [31] Ongaro, D., Ousterhout, J., 2014. In search of an understandable consensus algorithm, in: Proceedings of the 2014 USENIX Annual Technical Conference (USENIX ATC 14), pp. 305–319.
- [32] Ouaddah, A., Abou Elkalam, A., Ait Ouahman, A., 2016. FairAccess: A new blockchain-based access control framework for the Internet of Things. *Security and Communication Networks* 9, 5943–5964. doi:<https://doi.org/10.1002/sec.1748>.
- [33] Pajooh, H.H., Hammoudeh, M., Alam, F., Brocklehurst, S., 2021. Hyperledger fabric blockchain for securing the edge IoT: Systems architecture, experiments and critical observations. *Electronics* 10, 1338. doi:<https://doi.org/10.3390/electronics10111338>.
- [34] Putra, G.D., Dedeoglu, V., Kanhere, S.S., Jurdak, R., 2021. Trust management in decentralized IoT access control system, in: Proc. IEEE BCCA, pp. 1–8. doi:<https://doi.org/10.1109/BCCA53669.2021.9657049>.
- [35] Raza, S., Shafagh, H., Hewage, K., Hummen, R., Voigt, T., 2013. Lite: Lightweight secure CoAP for the Internet of Things. *IEEE Sensors Journal* 13, 3711–3720. doi:<https://doi.org/10.1109/JSEN.2013.2277656>.
- [36] Sandhu, R., Bhamidipati, V., Munawer, Q., 1999. The ARBAC97 model for role-based administration of roles. *ACM Transactions on Information and System Security* 2, 105–135. doi:<https://doi.org/10.1145/300830.300839>.
- [37] Sandhu, R.S., Coyne, E.J., Feinstein, H.L., Youman, C.E., 1996. Role-based access control models. *Computer* 29, 38–47. doi:<https://doi.org/10.1109/2.485845>.
- [38] Sciancalepore, S., Piro, G., Boggia, G., Bianchi, G., 2017. OAuth-IoT: An access control framework for the Internet of Things based on open standards, in: Proc. IEEE ISCC, pp. 676–681. doi:<https://doi.org/10.1109/ISCC.2017.8024606>.
- [39] Servos, D., Osborn, S.L., 2017. Current research and open problems in attribute-based access control. *ACM Computing Surveys* 49, 65:1–65:45. doi:<https://doi.org/10.1145/2983429>.
- [40] Shih, D.H., Jheng, P.Y., Yen, D.C., Huang, C.L., 2022. A blockchain-based access control model for industrial IoT. *Sensors* 22, 2188. doi:<https://doi.org/10.3390/s22062188>.
- [41] Tara, S., Zeadally, S., Sookun, D., 2023. Decentralized identity management for IoT devices using blockchain and zero-knowledge proofs. *IEEE Internet of Things Journal* 10, 3817–3827. doi:<https://doi.org/10.1109/JIOT.2022.3211254>.
- [42] Thakkar, P., Nathan, S., Viswanathan, B., 2018. Performance benchmarking and optimizing hyperledger fabric blockchain platform, in: Proceedings of the 26th IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS), pp. 264–276. doi:<https://doi.org/10.1109/MASCOTS.2018.00060>.
- [43] Tzenetopoulos, A., Katsaros, P., Mavridis, I., 2022. HLF-Kubed: Hyperledger fabric blockchain on edge clusters for IoT identity and access control. *Future Generation Computer Systems* 135, 271–283. doi:<https://doi.org/10.1016/j.future.2022.05.003>.
- [44] United States Department of Agriculture, 2024. National organic program (NOP) regulations. 7 CFR Part 205. Available at <https://www.ams.usda.gov/rules-regulations/organic>.
- [45] Usman, M., Jolfaei, A., Tariq, M.U., 2024. Blockchain-based access control for industrial IoT: Domain policy architecture with smart contracts. *Computer Networks* 243, 110286. doi:<https://doi.org/10.1016/j.comnet.2024.110286>.
- [46] Vetter, J., Novák, P., Wagner, M.R., Veit, C., Möller, K.C., Besenhard, J.O., Winter, M., Wohlfahrt-Mehrens, M., Vogler, C., Hammouche, A., 2005. Ageing mechanisms in lithium-ion batteries. *Journal of Power Sources* 147, 269–281. doi:<https://doi.org/10.1016/j.jpowsour.2005.01.006>.
- [47] Wang, N., Zhang, N., Wang, M., 2006. Wireless sensors in agriculture and food industry: Recent development and future perspective. *Computers and Electronics in Agriculture* 50, 1–14. doi:<https://doi.org/10.1016/j.compag.2005.09.003>.
- [48] Wang, X., Han, Y., Leung, V.C.M., Niyato, D., Yan, X., Chen, X., 2020. Convergence of edge computing and deep learning: A comprehensive survey. *IEEE Communications Surveys & Tutorials* 22, 869–904. doi:<https://doi.org/10.1109/COMST.2020.2970550>.
- [49] World Wide Web Consortium, 2022. Decentralized identifiers (DIDs) v1.0. W3C Recommendation, 19 July 2022. Available at <https://www.w3.org/TR/did-core/>.
- [50] World Wide Web Consortium, 2024. Bitstring status list v1.0. W3C Candidate Recommendation Snapshot, 5 March 2024. Available at <https://www.w3.org/TR/vc-bitstring-status-list/>.
- [51] Xu, R., Chen, Y., Blasch, E., Chen, G., 2018. BlendCAC: A blockchain-enabled decentralized capability-based access control for IoTs, in: Proceedings of the 2018 IEEE International Conference on Internet of Things (iThings), pp. 1027–1034. doi:https://doi.org/10.1109/Cybermatics_2018.2018.00191.
- [52] Ye, N., Zhu, Y., Wang, R., Malekian, R., Lin, Q., 2014. An efficient authentication and access control scheme for perception layer of internet of things. *Applied Mathematics & Information Sciences* 8, 1617–1624. doi:<https://doi.org/10.12785/amis/080416>.
- [53] Yin, M., Malkhi, D., Reiter, M.K., Gueta, G.G., Abraham, I., 2019. HotStuff: BFT consensus with linearity and responsiveness, in: Proceedings of the 2019 ACM Symposium on Principles of Distributed Computing (PODC), pp. 347–356. doi:<https://doi.org/10.1145/3293611.3331591>.
- [54] Zaidi, S.M.S., Al-Qurishi, M., Abidin, M.Z., Almakdi, S., Asiri, Y., 2023. Fabrication: A hyperledger fabric framework for scalable IoT access control. *IEEE Internet of Things Journal* 10, 12636–12649. doi:<https://doi.org/10.1109/JIOT.2023.3254978>.

- [55] Zhang, N., Wang, M., Wang, N., 2002. Precision agriculture: A worldwide overview. *Computers and Electronics in Agriculture* 36, 113–132. doi:[https://doi.org/10.1016/S0168-1699\(02\)00056-X](https://doi.org/10.1016/S0168-1699(02)00056-X).